Wang 2019 Glioblastoma — analysis summary

RUN 3 OF 8

GO / NO-GO GATES

Gate 1 — mixing & continuity → GO

Samples intermix on one continuous connected manifold. Per-neighbourhood iLISI rises 1.03 (pre-Harmony) → 2.60 (θ=10) and plateaus; n_pcs 20/40 near-identical (not under-powered). One component after island removal.

Gate 2 — module geography → GO (non-branching)

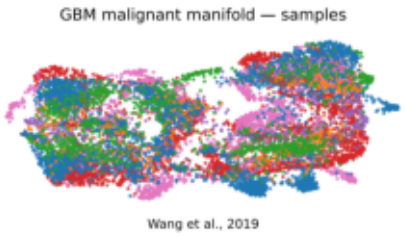
Embryonic module pattern present (non-branching). Gast/pSom co-vary spatially (r=+0.16) along one axis and both oppose Crest (Gast r=−0.49, pSom r=−0.87); Crest is the inverse terminal gradient (high-decile pseudotime 0.84).

Gate 3 — hub cross-repression → NA

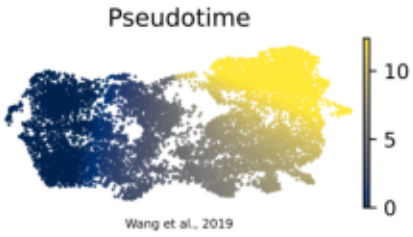
Not applicable — no bifurcation. Gast and pSom co-vary rather than cross-repress, so there is no Gast/pSom hub diverging into two competing arms. Geometry is linear (curve: 2 tips, 0 forks, fold-free ratio 2.82).

STEP FIGURES

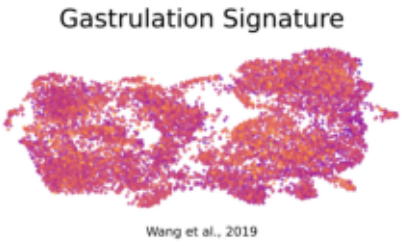
Manifold — samples



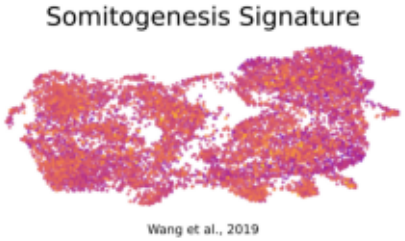
Pseudotime



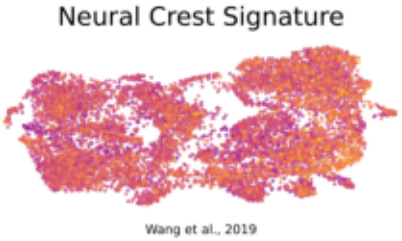
Gastrulation



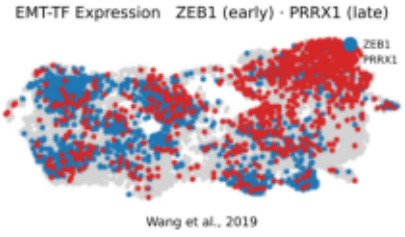
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
Glioblastoma (Wang et al., 2019), 10x — 9,632 malignant cells, 7 samples.

Verdict.
The embryonic program reproduces as a single collapsed (linear) axis: a Gast/pSom-high developmental origin grading to a Crest-high terminus, EMT-TFs ZEB1 (early) → PRRX1 (late). No Gast/pSom bifurcation → Gate 3 NA. Gate 1 GO / Gate 2 GO (non-branching). Module means Gast −0.002 / pSom +0.008 / Crest −0.007.

Parameter choices

Min-cell threshold	237 (7/10 samples; 9,650 cells); largest ratio gap 396→143 (2.77), cut at geom mean
Island removal	UMAP radius graph (6× median NN), keep largest component → 18 cells / 2 specks → 9,632
Geometry / root	Linear non-branching scFates curve; root = Gast/pSom-high, Crest-low tip; continuity +0.98 (diffusion)
Integration	Harmony θ=10, n_pcs=20; iLISI plateaus 2.4→2.6 (θ 5→10); no sc.pp.scale
Subtypes	AC-like / OPC-like / NPC-like marker scores per Leiden cluster (publication meta-programs)
Module means (raw)	Gast −0.002 / pSom +0.008 / Crest −0.007 (Linear-pSom/Crest)